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Title: Detection of Microsleep Events With a Behind-the-Ear Wearable System

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Contributors: Pham, N.; Dinh, T.; Kim, T.; Raghebi, Z.; Bui, N.; Truong, H.; Nguyen, T.; Banaei-Kashani, F.; Halbower, A.; Nguyen, P.; Vu, T.;

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Abstract: Every year, the U.S. economy loses more than \$411 billion because of work performance reduction, injuries, and traffic accidents caused by microsleep. To mitigate microsleep's consequences, an unobtrusive, reliable, and socially acceptable microsleep detection solution throughout the day, every day is required. Unfortunately, existing solutions do not meet these requirements. In this paper, we propose WAKE, a novel behind-the-ear wearable device for microsleep detection. By monitoring biosignals from the brain, eye movements, facial muscle contractions, and sweat gland activities from behind the user's ears, WAKE can detect microsleep with a high temporal resolution. We introduce a Three-fold Cascaded Amplifying (3CA) technique to tame the motion artifacts and environmental noises for capturing high fidelity signals. Through our prototyping, we show that WAKE can suppress motion and environmental noise in real-time by 9.74-19.47 dB while walking, driving, or staying in different environments, ensuring that the biosignals are captured reliably. We evaluated WAKE using gold-standard devices on 19 sleep-deprived and narcoleptic subjects. The Leave-One-Subject-Out Cross-Validation results show the feasibility of WAKE in microsleep detection on an unseen subject with average precision and recall of 76 and 85 percent, respectively.

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